

Computer Networks Assignment

Q1. Characteristics of a Computer Network

A computer network is a group of two or more computers and devices connected together to share information, resources, and services. For example, in a college computer lab, all computers may be connected to one network so students can access the Internet and use the same printer.

Resource sharing: Users can share hardware such as printers, scanners, storage devices, and Internet connections. For example, 20 computers can use one network printer.

Communication: Networks allow users to communicate through email, messaging, video calls, and file sharing. For example, teachers can send notes or assignments to students through the college network.

Data sharing: Files, documents, databases, and software can be accessed by authorized users from different computers. For example, office staff can access student records stored on a central server.

Reliability: Important data can be stored on multiple systems or backed up on servers. If one computer fails, data may still be available from another device or backup.

Scalability: A network can be expanded by adding more computers, printers, or other devices when needed. For example, a lab with 30 computers can later add 20 more computers.

Security: Networks use passwords, firewalls, and access control to protect data from unauthorized users. For example, only teachers may be allowed to modify examination records.

Q2. Goals, Advantages, and Disadvantages of Computer Networks

The main goal of a computer network is to connect computers and devices so users can communicate, share resources, and access information easily. Networks help organizations store data centrally and allow authorized users to access it from different locations.

Efficient communication: Networks provide email, messaging, video conferencing, and file transfer facilities. For example, teachers and students can communicate through email or online classes.

Resource sharing: Hardware and software can be shared among many users. For example, several computers in a lab can use one printer and one Internet connection.

Centralized data management: Important files and records can be stored on a central server. For example, student records can be managed from one database.

Improved collaboration: Users can work together on shared documents and projects. For example, office staff can update the same report from different computers.

The advantages are lower cost, better productivity, remote access, and easy expansion. However, networks also have disadvantages such as security risks, viruses, hacking, setup cost, maintenance requirements, and failure of central devices.

Q3. Classification of Computer Networks

Computer networks are classified according to the geographical area they cover. The three main types are LAN, MAN, and WAN.

LAN (Local Area Network): A LAN connects computers within a small area such as a home, office, school, laboratory, or college campus. For example, the computers in a college laboratory connected

to a switch form a LAN.

MAN (Metropolitan Area Network): A MAN covers a larger area than a LAN but a smaller area than a WAN. It connects different LANs within a city or metropolitan area. For example, a university may connect its different campuses located in the same city through a MAN.

WAN (Wide Area Network): A WAN covers a very large geographical area, such as a state, country, or the whole world. It connects many LANs and MANs. For example, the Internet is the largest WAN because it connects computers and networks worldwide.

Q4. Comparison of Network Topologies

A network topology is the physical or logical arrangement of computers, cables, and other devices in a network. Four common topologies are star, bus, ring, and mesh.

Star topology: Every computer is connected to a central hub or switch. It is easy to install, manage, and expand. If one cable fails, only one computer is affected. However, if the central hub fails, the whole network stops working. It is commonly used in schools, offices, and computer labs.

Bus topology: All computers are connected to one main cable. It is low cost and suitable for small networks. However, if the main cable fails, the complete network may stop. It was commonly used in early LANs.

Ring topology: Each computer is connected to two other computers, forming a circular path. Data moves in one direction around the ring. It has controlled data transmission, but a cable or device failure can affect the whole network.

Mesh topology: Every device has direct connections with other devices. It is highly reliable and fast, but expensive because it needs more cables and devices. It is used in important WAN and communication networks.

Q5. Role of Networking Standard Organizations

Networking standard organizations develop rules and standards so that computers, devices, and networks from different companies can communicate properly.

ISO (International Organization for Standardization): ISO develops international standards for networking and other technologies. Its important contribution is the OSI Model, which explains how network communication works in layers.

IEEE (Institute of Electrical and Electronics Engineers): IEEE creates standards for LANs and wireless networks. For example, IEEE 802.3 is used for Ethernet networks and IEEE 802.11 is used for Wi-Fi networks.

IETF (Internet Engineering Task Force): IETF develops and maintains Internet standards and protocols. Important Internet protocols such as TCP and IP are developed through IETF standards.

ITU (International Telecommunication Union): ITU sets global standards for telecommunication systems. It helps different countries and companies use compatible communication technologies.

ICANN (Internet Corporation for Assigned Names and Numbers): ICANN manages domain names and IP address allocation. It supports the Domain Name System, which converts website names into IP addresses.

Q6. Recommended Network for a College Computer Laboratory

For a college laboratory with 50 computers, shared printers, Wi-Fi, and Internet access, a **LAN using star topology** is the most suitable choice.

In this setup, all computers should be connected to a central network switch using Ethernet cables. Shared printers and Wi-Fi access points can also be connected to the same switch. The switch should connect to a router, which provides Internet access to all computers.

Star topology is suitable because it is easy to install, manage, and expand. If one computer or its cable fails, the other computers continue to work normally. This is very useful in a computer lab because one faulty system should not stop the entire network.

The network can also be expanded later by connecting more computers or printers to additional switch ports. Wi-Fi access points will allow teachers and students to connect laptops and mobile devices. Therefore, a LAN with star topology provides good speed, reliability, security, and easy maintenance for the college laboratory.